NAME:____________________________ **DUE MAY 11, 2026, 4:00**

FINAL TAKEHOME EXAM

ESC201M, MECHANICS OF MATERIALS, COOPER UNION

Ground Rules:

By submitting this exam solution with your name, you affirm that you have abided by the code of conduct of Cooper Union. You affirm that this exam solution is your own work, and that you did not receive unauthorized assistance from any person.

Please address all questions about this exam to the instructor.

This exam is open book(s) and open notes, and calculators and computers are allowed. Your exam should be your own work, and this material will be the basis for the in-class final exam. Discuss any questions about the exam and project with the instructor via Teams chat, the Teams office hours/discussion channel, or email. You can reach the instructor via text message (267 997 7383) for urgent questions.

Bring a hard copy of your exam to the in-class final exam. You may refer to this exam during the in-class exam. You will turn in both exams at the end of the in-class exam period.

You may upload supporting calculation tools (Excel workbook, MATLAB and/or Python scripts) to MS Teams, but I will be mostly relying on the exam hardcopy that you turn in for clear answers.

The Big Picture: You will analyze and explore several different aspects of the airframe design of the 2026 Vertical Flight Society (VFS) Vertical Take-Off and Landing (VTOL aircraft. This VTOL (Figure 1) is an unconventional quadcopter designed to pick up, transport, and drop bags of fire-fighting material to combat wildfires.

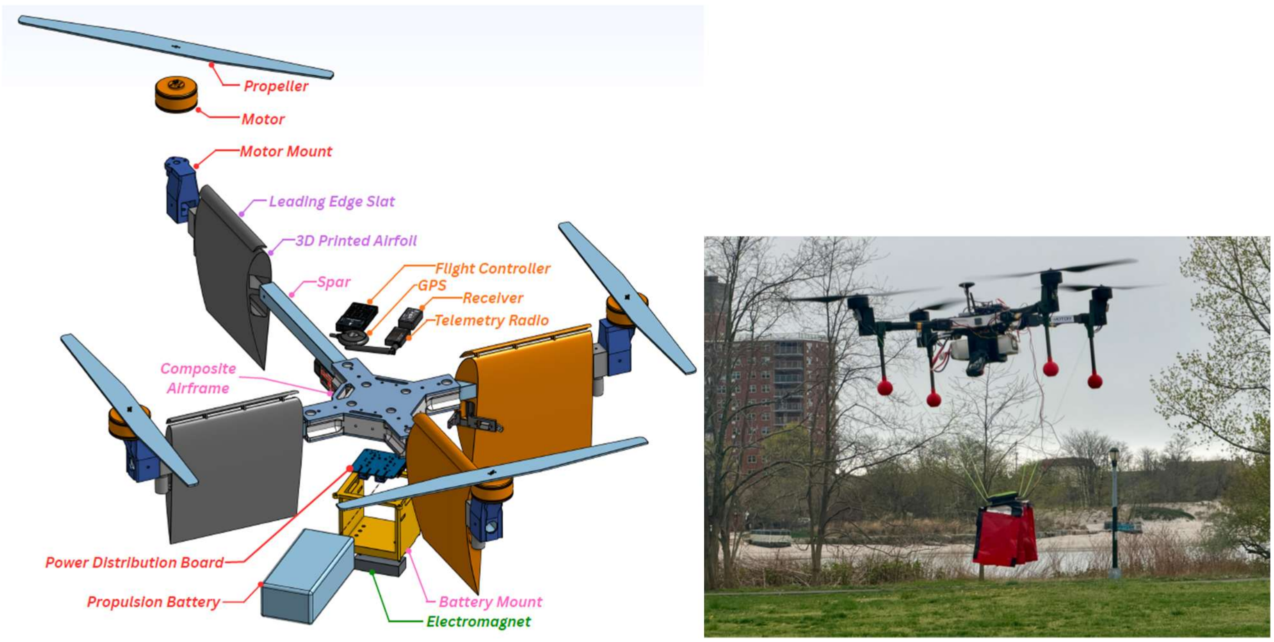

Figure 1. VFS VTOL aircraft (credit: 2025-2026 VTOL Team). (Left) Exploded view of most major components, showing three assembled spar-wing-motor-propellor assembles and one in exploded view, along with the central composite airframe body. (Right) VTOL flying with payload and legs (not shown in left figure) and with airfoils removed.

Key specs for the aircraft include the loaded weight (89 N), the maximum thrust per propellor (80 N), and the estimated maximum lift and drag per airfoil (16N and 3N respectively).

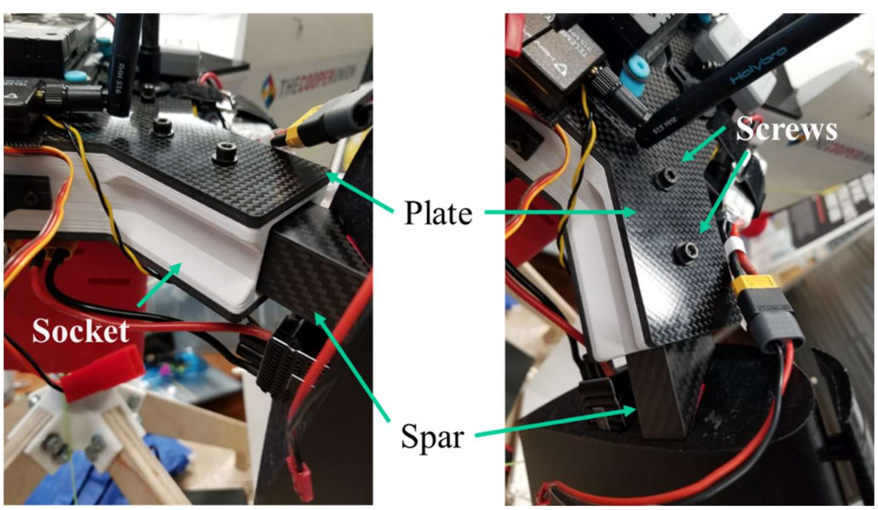

Figure 2. Central airframe hub socket closeup images. (Left) Side oblique view highlighting the hollow I-beam socket, plate, and base of the wing spar. Note 0°/90° carbon fiber weave in spar. (Right) Top oblique view highlighting the top mounting screws. Note ±45° weave orientation in top plate.

The composite airframe at the center of the VTOL consists of a 3D printed hub sandwiched between two thin plates. The 3D printed hub has four sockets that hold the wing spars in position and each plate, socket, and spar is held together by two pairs of 3mm screws (2 on top and 2 on bottom) as shown in Figure 2. The plates are 4mm thick carbon fiber epoxy composite plates. To maximize the span of the plate, it was machined so that the carbon fibers are oriented at ±45° relative to the spar axis. The 3D printed socket is manufactured from solid infill ASA, in the shape of a box beam with lateral extensions and triangular gussets (Figure 3).

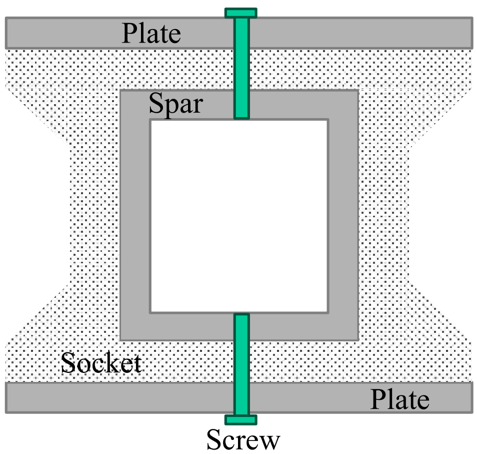

Figure 3. Airframe socket cross-section, current VTOL design.

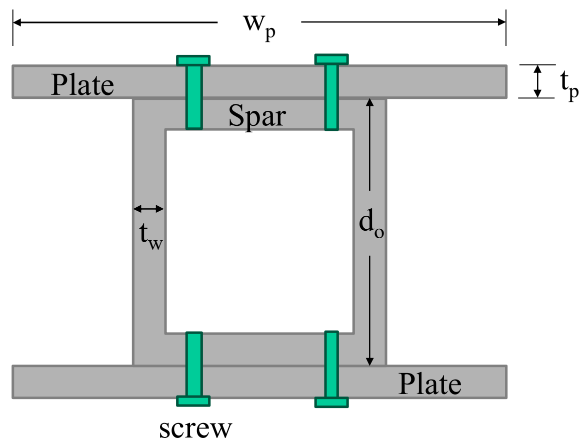

Figure 4. Simplified airframe/spar fixture cross-section without 3DP ASA socket.

The 3D printed body gives wide design flexibility for enclosing and cradling components, but isn't weatherproof and may not be weight-efficient, so the project advisor would like you to analyze a simpler structure, retaining the two carbon fiber plates and attaching them to the carbon fiber spars using four pairs of screws instead of two pairs. The cross-section of the new arrangement is shown in Figure 4. We will also buy a larger plate with the carbon fibers oriented at 0°/90°, so that the modulus and strength of the plate and spar will be similar.

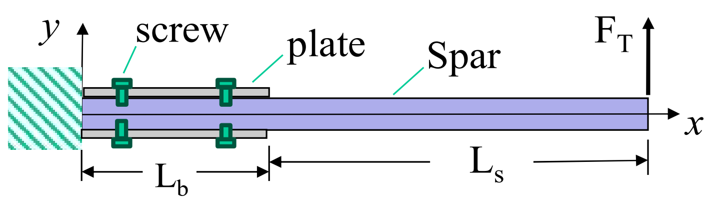

Figure 5. Side view of spar with applied thrust for F_T. Plate length overlapping with the spar is L_b, and free spar length is L_s. Each plate is connected to the spar with 2 pairs of screws.

Problem 1: (50 pts) Based on past experience (the 2024 VTOL was unstable due to low-frequency high-amplitude vibration) we are very concerned with the stiffness of the spar at the motor, where the propellor thrust is applied. Estimate the deflection of one spar when maximum thrust force (F_T = 80 N) is applied to the tip of the spar. You may assume that the center of the airframe is rigid, allowing zero rotation and zero deflection, but that the plate/spar joint is deformable and acts as a short composite beam. You may ignore the effect of the airfoils (lift, drag, stiffness) for this model.

a. Derive expressions for the slope and deflection of the spar/plate system illustrated in Figure 5, at location $x = L_b$, and at the tip of the spar ($x = L_b + L_s$), in terms of the dimensions of the plates and spar, and the applied thrust force. Clearly define any convenient intermediate terms or variables that you use.

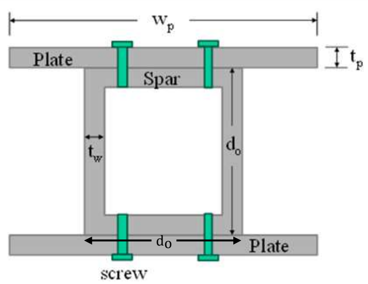

Figure 4. Simplified airframe/spar fixture cross-section without 3DP ASA socket.

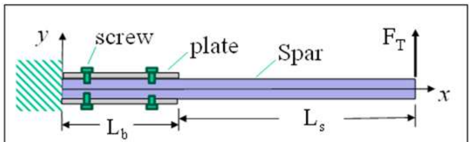

Figure 5. Side view of spar with applied thrust for F_T. Plate length overlapping with the spar is L_b, and free spar length is L_s. Each plate is connected to the spar with 2 pairs of screws.

1

Given:

- $F_T = 80\ N$
- The modulus of elasticity and ultimate strength of the plate and spar is similar:
  - $E = E_{spar} = E_{plate}$, $\sigma_{ult} = \sigma_{ult,spar} = \sigma_{ult,plate}$
- Let the total length be $L_T = L_b + L_s$

Approach:

- Since the material properties of the plate and spar are approximately equal, I can treat them as a single composite beam and solve for the resulting deflection.
- Note that while the material properties are constant throughout the beam, the area moment of inertia (I) is discontinuous at $x = L_b$.
- First, deriving an internal shear and moment distribution using section FBDs of the beam.
- Then, I can solve the ODE to find the beam deflections and slopes:

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

- Accounting for the discontinuity in I, I will solve the ODE for two regions: $(0 < x < L_b)$ and $(L_b < x < L_T)$. Using initial conditions at $x = L_b$ to link the two regions:

$$y_{0\to L_b}(L_b) = y_{L_b\to L_T}(L_b)$$
$$\theta_{0\to L_b}(L_b) = \theta_{L_b\to L_T}(L_b)$$

Where $\theta(x) = y'(x)$

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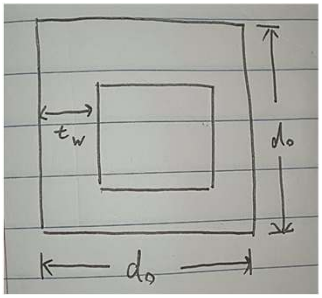
Geometry

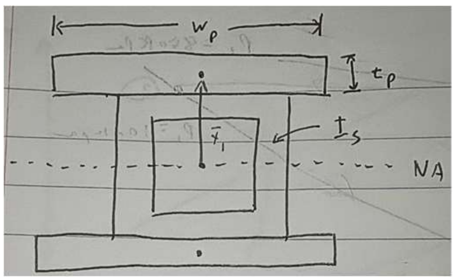
Area Moment of Inertia of just the Spar I_s by Superposition:



$$I_s = \frac{1}{12}(d_o)^4 - \frac{1}{12}(d_o - 2t_w)^4$$

$$\boxed{I_s = \frac{1}{12}[d_o^4 - (d_o - 2t_w)^4]}$$

Area Moment of Inertia of just the Spar and Plate Composite I_{s+p} by Parallel Axis Theorem:



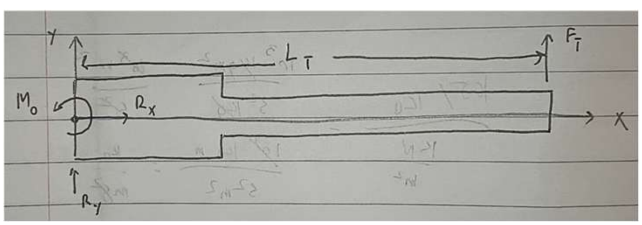
$$I_{p,NA} = \frac{1}{12}w_p t_p^3 + [w_p t_p]\bar{y}_1^2, \qquad \bar{y}_1 = \frac{d_o}{2} + \frac{t_p}{2}$$

$$I_{p,NA} = \frac{1}{12}w_p t_p^3 + [w_p t_p]\left(\frac{d_o + t_p}{2}\right)^2$$

$$I_{s+p} = I_s + 2I_{p,NA}$$

$$\boxed{I_{s+p} = \frac{1}{12}[d_o^4 - (d_o - 2t_w)^4] + 2\left[\frac{1}{12}w_p t_p^3 + [w_p t_p]\left(\frac{d_o + t_p}{2}\right)^2\right]}$$

FBDs

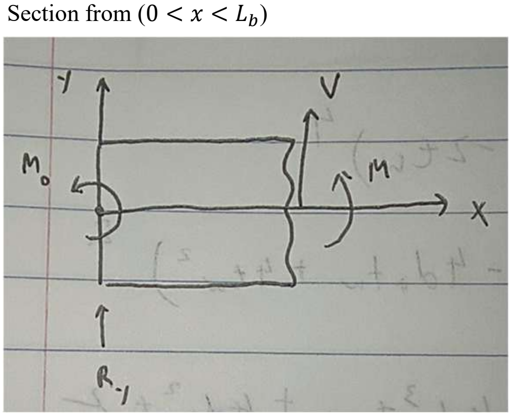


$$\sum F_x = 0 = R_x \rightarrow R_x = 0$$

$$\sum F_y = 0 = F_T + R_y \rightarrow R_y = -\mathrm{F_T}$$

$$\sum M_{z,0} = 0 = M_0 + F_T(L_T) \rightarrow M_0 = -\mathrm{F_T L_T}$$

Section from $(0 < x < L_b)$

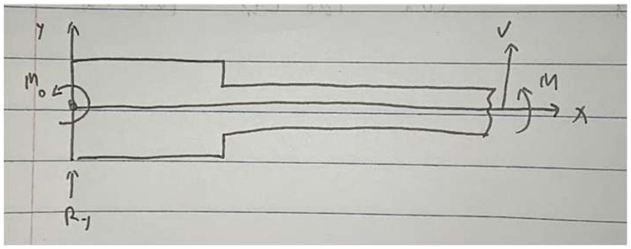


$$\sum F_y = 0 = V + R_y \rightarrow V = -\mathrm{R_y} \qquad \boxed{V_{0\rightarrow L_b}(x) = F_T}$$

$$\sum M_{z,0} = 0 = M_0 + M + Vx \rightarrow M = -M_0 - Vx$$

$$\boxed{M_{0\rightarrow L_b}(x) = F_T(L_T - x)}$$

Section from $(L_b < x < L_T)$

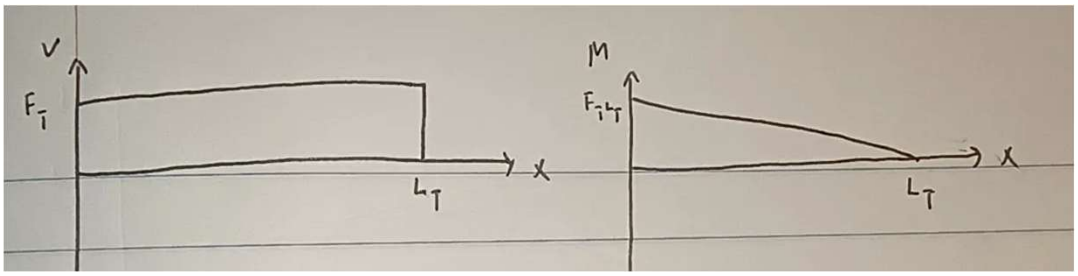


$$\sum F_y = 0 = V + R_y \;\rightarrow\; V = -\mathrm{R_y} \qquad \boxed{V_{L_b \to L_T}(x) = F_T}$$

$$\sum M_{z,0} = 0 = M_0 + M + Vx \rightarrow \; M = -M_0 - Vx$$

$$\boxed{M_{L_b \to L_T}(x) = F_T(L_T - x)}$$

Moment and Shear Load Distribution:



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Solving the ODE:

Section from $(0 < x < L_b)$

$$\frac{d^2 y_{0\to L_b}}{dx^2} = \frac{M(x)}{EI_{s+p}} = \frac{1}{EI_{s+p}}\left[F_T(L_T - x)\right]$$

Initial Conditions: $y_{0\to L_b}(0) = 0,\; \theta_{0\to L_b}(0) = 0$ (Fixed Support)

$$\theta_{0\to L_b} = \frac{dy_{0\to L_b}}{dx} = \int_0^x \frac{1}{EI_{s+p}}\left[F_T(L_T - x')\right]dx' + \theta_{0\to L_b}(0)$$

$$\boxed{\theta_{0\to L_b} = \frac{F_T}{EI_{s+p}}\left[L_T x - \frac{x^2}{2}\right]}$$

$$y_{0\to L_b} = \int_0^x \frac{F_T}{EI_{s+p}}\left[L_T x' - \frac{x'^2}{2}\right]dx' + y_{0\to L_b}(0)$$

$$\boxed{y_{0\to L_b} = \frac{F_T}{EI_{s+p}}\left[\frac{L_T x^2}{2} - \frac{x^3}{6}\right]}$$

Solve for Deflection and Slope at $x = L_b$ (Initial Conditions for Section $(L_b < x < L_T)$)

$$\theta_{0\to L_b}(L_b) = \frac{F_T}{EI_{s+p}}\left[L_T L_b - \frac{{L_b}^2}{2}\right] = \frac{F_T L_b}{EI_{s+p}}\left[L_T - \frac{L_b}{2}\right] \qquad \boxed{\theta_{0\to L_b}(L_b) = \frac{F_T L_b}{2EI_{s+p}}\left[2L_T - L_b\right]}$$

$$y_{0\to L_b}(L_b) = \frac{F_T}{EI_{s+p}}\left[\frac{L_T {L_b}^2}{2} - \frac{{L_b}^3}{6}\right] = \frac{F_T L_b^2}{EI_{s+p}}\left[\frac{L_T}{2} - \frac{L_b}{6}\right] \qquad \boxed{y_{0\to L_b}(L_b) = \frac{F_T L_b^2}{6EI_{s+p}}\left[3L_T - L_b\right]}$$

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Section from ($L_b < x < L_T$)

$$\frac{d^2 y_{L_b\to L_T}}{dx^2} = \frac{M(x)}{EI_s} = \frac{1}{EI_s}[F_T(L_T - x)]$$

Initial Conditions: $y_{L_b\to L_T}(L_b) = y_{0\to L_b}(L_b)$, $\theta_{L_b\to L_T}(L_b) = \theta_{0\to L_b}(L_b)$

$$\theta_{L_b\to L_T} = \frac{dy_{L_b\to L_T}}{dx} = \int_{L_b}^{x} \frac{1}{EI_s}[F_T(L_T - x')]\,dx' + \theta_{L_b\to L_T}(L_b)$$

$$\theta_{L_b\to L_T} = \frac{F_T}{EI_s}\left(L_T x' - \frac{x'^2}{2}\right)\Bigg|_{L_b}^{x} + \frac{F_T L_b}{2EI_{s+p}}[2L_T - L_b]$$

$$\theta_{L_b\to L_T} = \frac{F_T}{EI_s}\left(L_T x - \frac{x^2}{2}\right) - \frac{F_T L_b}{2EI_s}(2L_T - L_b) + \frac{F_T L_b}{2EI_{s+p}}[2L_T - L_b]$$

$$\boxed{\theta_{L_b\to L_T} = \frac{F_T}{2EI_s}(2L_T x - x^2) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b]}$$

Slope at $x = L_T$

$$\theta_{L_b\to L_T}(L_T) = \frac{F_T}{2EI_s}(2L_T^2 - L_T^2) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b]$$

$$\boxed{\theta_{L_b\to L_T}(L_T) = \frac{F_T L_T^2}{2EI_s} + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b]}$$

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$$y_{L_b\to L_T} = \int_{L_b}^{x}\left(\frac{F_T}{EI_s}\left(L_T x' - \frac{x'^2}{2}\right) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b]\right)dx' + y_{L_b\to L_T}(L_b)$$

$$y_{L_b\to L_T} = \frac{F_T}{EI_s}\left(\frac{L_T x'^2}{2} - \frac{x'^3}{6}\right)\Bigg|_{L_b}^{x} + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](x - L_b) + y_{L_b\to L_T}(L_b)$$

$$y_{L_b\to L_T} = \frac{F_T}{6EI_s}(3L_T x^2 - x^3) - \frac{F_T L_b^2}{6EI_s}(3L_T - L_b) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](x - L_b) + \frac{F_T L_b^2}{6EI_{s+p}}[3L_T - L_b]$$

$$\boxed{y_{L_b\to L_T} = \frac{F_T}{6EI_s}(3L_T x^2 - x^3) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](x - L_b) + \left(\frac{F_T L_b^2}{6EI_{s+p}} - \frac{F_T L_b^2}{6EI_s}\right)[3L_T - L_b]}$$

Deflection at $x = L_T$

$$y_{L_b\to L_T}(L_T) = \frac{F_T}{6EI_s}(3L_T^3 - L_T^3) + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](L_T - L_b) + \left(\frac{F_T L_b^2}{6EI_{s+p}} - \frac{F_T L_b^2}{6EI_s}\right)[3L_T - L_b]$$

$$\boxed{y_{L_b\to L_T}(L_T) = \frac{F_T L_T^3}{3EI_s} + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](L_T - L_b) + \left(\frac{F_T L_b^2}{6EI_{s+p}} - \frac{F_T L_b^2}{6EI_s}\right)[3L_T - L_b]}$$

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Answers to Problem 1a:

- Slope at $x = L_b$

$$\theta_{0\to L_b}(L_b) = \frac{F_T L_b}{2EI_{s+p}}[2L_T - L_b]$$

- Deflection at $x = L_b$

$$y_{0\to L_b}(L_b) = \frac{F_T L_b^2}{6EI_{s+p}}[3L_T - L_b]$$

- Slope at $x = L_T$

$$\theta_{L_b\to L_T}(L_T) = \frac{F_T L_T^2}{2EI_s} + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b]$$

- Deflection at $x = L_T$

$$y_{L_b\to L_T}(L_T) = \frac{F_T L_T^3}{3EI_s} + \left(\frac{F_T L_b}{2EI_{s+p}} - \frac{F_T L_b}{2EI_s}\right)[2L_T - L_b](L_T - L_b) + \left(\frac{F_T L_b^2}{6EI_{s+p}} - \frac{F_T L_b^2}{6EI_s}\right)[3L_T - L_b]$$

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b. Calculate the slope (θ) and deflection (y) at $x = L_b$ and $x = L_b + L_s$ for the dimensions and properties given in Table 1

t_p	4	mm
w_p	44	mm
t_w	2	mm
d_o	25	mm
L_b	76	mm
L_S	400	mm
$E_{CF,0}$	40	GPa
$E_{CF,45}$	10	GPa
E_{ASA}	2.1	GPa
$\sigma_{u,0}$	800	Mpa

Table 1. Dimensions and properties of plate and wing spar. (Use $E_{CF,0}$ and $\sigma_{u,0}$ for modulus and strength, respectively, of 0°/90° carbon fiber.)

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Geometry

$$I_s = \frac{1}{12}[d_o^4 - (d_o - 2t_w)^4]$$

$$I_s = \frac{1}{12}\left[[0.025\,m]^4 - (0.025 - 2 \cdot 0.002\,m)^4\right] \qquad \boxed{I_s = 1.634 \times 10^{-8}\,m^4}$$

$$I_{s+p} = \frac{1}{12}[d_o^4 - (d_o - 2t_w)^4] + 2\left[\frac{1}{12}w_p t_p^3 + [w_p t_p]\left(\frac{d_o + t_p}{2}\right)^2\right]$$

$$I_{s+p} = [1.634 \times 10^{-8}\,m^4] + 2\left[\frac{1}{12}[0.044\,m][0.004\,m]^3 + [0.044\,m][0.004\,m]\left(\frac{[0.025 + 0.004\,m]}{2}\right)^2\right]$$

$$\boxed{I_{s+p} = 9.081 \times 10^{-8}\,m^4}$$

$$\boxed{L_T = L_b + L_s = [0.076 + 0.4\,m] = 0.476\,m}$$

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Using a Python Script to Plug in the Parameters:

```
# --- Moment of Inertia Calculations ---

# Spar only (Hollow Box)
I_s = (1/12)*(d_o**4 -(d_o - 2*t_w)**4)  # [m^4]

# Combined section (Spar + 2 Plates)
I_sp = I_s + 2 * ((1/12)*w_p*t_p**3+w_p*t_p*(d_o/2 + t_p/2)**2)  # [m^4]

# --- Slope and Deflection Calculations  ---
## Slope at x=L_b
theta_Lb = ((F_T*L_b)/(E*I_sp))*(2*L_T-L_b)  # [radians]
## Deflection at x=L_b
y_Lb = ((F_T*L_b**2)/(6*E*I_sp))*(3*L_T-L_b)  # [m]
## Slope at x=L_T
theta_LT = (
((F_T*L_T**2)/(2*E*I_s))+
(((F_T*L_b)/(2*E*I_sp))-((F_T*L_b)/(2*E*I_s)))*(2*L_T-L_b)  # [radians]
)
## Deflection at x=L_T
y_LT = (
((F_T*L_T**3)/(3*E*I_s))+
(((F_T*L_b)/(2*E*I_sp))-((F_T*L_b)/(2*E*I_s)))*(2*L_T-L_b)*(L_T-L_b)+
(((F_T*L_b**2)/(6*E*I_sp))-((F_T*L_b**2)/(6*E*I_s)))*(3*L_T-L_b)
)
```

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Answers to Problem 1b:

```
I_spar: 1.6345e-08 m^4
I_composite: 9.0823e-08 m^4

--- Results at x = Lb (76.0 mm) ---
Slope: 0.001466 rad
Deflection: 0.0287 mm

--- Results at Tip (x = 476.00000000000006 mm) ---
Slope: 0.010522 rad
Deflection: 2.9322 mm
```

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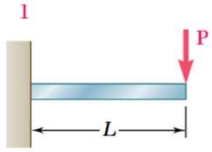
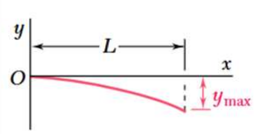
c. Check your work by doing two very simple (back of the book) calculations that should bound your result.

Approach:

- The total deflection at the tip $x = L_T$ must be bounded between the following two cases:
  - 1) The entire beam has the area moment of inertia I_{s+p}, giving us the *minimum* deflection case as it is stiffer
  - 2) The entire beam has the area moment of inertia I_s, giving us the *maximum* deflection case
- From the textbook, the maximum deflection for a cantilever beam is (note: for our loading, the max deflection will be positive)

Beam and Loading	Elastic Curve	Maximum Deflection
1 		$-\frac{PL^3}{3EI}$

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Using Python:

```
# Bounding cases:
# Minimum deflection: maximum stiffness (I_sp)
y_min = ((F_T*L_T**3)/(2*E*I_sp))  # [m]
# Maximum deflection: minimum stiffness (I_s)
y_max = ((F_T*L_T**3)/(2*E*I_s))  # [m]
```

Answers to Problem 1c:

Comparing the deflection calculated at the tip (2.9322 mm) to the bounding cases, it can be seen that it lies in between and is thus valid.

```
--- Bounding Cases ---
Minimum Deflection: 1.1875 mm
Maximum Deflection: 6.5982 mm
```

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d. The carbon fiber can be treated as a brittle material. Ignoring the screws for now, estimate the factor of safety at maximum thrust. (Clearly state the failure criterion that you are using.)

Approach:

- Since the carbon fiber is a brittle material, I will use **Mohr's Criterion:**
  - Failure (yielding) occurs when the maximum normal stress exceeds the ultimate strength σ_U
- The factor of safety is then defined by

$$F.S. = \frac{\sigma_U}{\sigma_{max}}$$

- Where the maximum normal stress in the beam is given by the bending stress:

$$\sigma_{max} = \frac{M_{max}c}{I}$$

Identify Max Moment

$$M_{max} = M(0) = F_T L_T$$

Beam Geometry at Max Moment

This is in section $(0 < x < L_b)$, thus $I = I_{s+p}$, and the distance from the upper face to the neutral axis c is:

$$c = \frac{d_o}{2} + t_p$$

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Solving for the Max Stress with Python

```
# Max stress:
sigma_max = ((F_T*L_T)*(d_o/2 + t_p))/I_sp  # [Pa]
# Factor of safety:
FOS = sigma_u/sigma_max
```

Answers to Problem 1d:

```
--- Factor of Safety ---
Maximum Stress: 6.92 MPa
Factor of Safety: 115.64
```

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f. Discuss your results. Do the deflection and factor of safety seem adequate?

Given that the length of the entire spar is 476 mm and the deflection at max thrust is only 2.93 mm, the design is very stiff. Since the goal of the design is to minimize low frequency instabilities, increasing the stiffness of the spar increases the natural frequency of the spar ($\omega_n = \sqrt{k/m}$), thus moving it away from any sort of resonant frequencies generated from the motors. Additionally, the factor of safety was extremely high (115.64), which means that the material of the design was chosen mostly for its stiffness properties as the design is currently very overengineered in terms of strength.

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Problem 2: (25 pts.) Now consider the loads on the screws. Estimate the shear force transmitted by each of the 8 screws that hold the plates and spar together. Assume that the screws bear equal shear loads. Discuss if the shear force is reasonable for a 3mm diameter screw.

Approach:

- As the spar bends, there will be horizontal shear load per unit length (shear flow q). At the section with the plates ($0 < x < L_b$), this horizontal shear load will be transmitted from the spar to the plate via the screws.
$$q = \frac{VQ}{I}\ \left[\frac{N}{m}\right]$$
  - $Q = Q_{plate} = \bar{y}_{plate}A_{plate}$ is the first area moment of the plate around the neutral axis
  - $I = I_{s+p}$
  - $V = F_T$
- To find the total shear load transmitted to the plates, I will multiply the shear flow by the length of the plate
$$F_{shear} = qL_b\ \ [N]$$
- Since there are 4 screws resisting the horizontal shear load at each plate. If all screws bear equal loads, the load per screw will be
$$F_{screw} = \frac{F_{shear}}{4}$$
- Finally, to discuss whether the shear force is reasonable for a 3mm screw, I will calculate the average shear stress for each screw and compare to yield strength of steel to see if failure will occur
$$\tau_{avg} = \frac{F_{screw}}{\frac{\pi}{4}d^2} = \frac{4F_{screw}}{\pi d^2}$$

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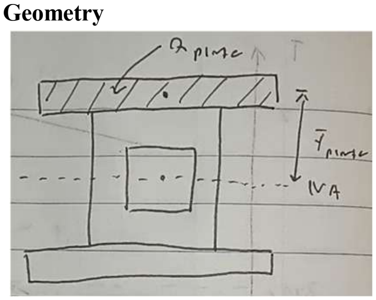
Geometry



$$\boxed{I_{s+p} = 9.081 \times 10^{-8}\ m^4}$$

$$\bar{y}_{plate} = \frac{d_o + t_p}{2}$$

$$Q_{plate} = \left(\frac{d_o + t_p}{2}\right)w_p t_p = \left(\frac{[0.025 + 0.004\ m]}{2}\right)[0.044\ \ m][0.004\ m]$$

$$\boxed{Q_{plate} = 2.552 \times 10^{-6}\ m^3}$$

Shear Flow

$$q = \frac{VQ_{plate}}{I_{s+p}} = \frac{[80\ N][2.552 \times 10^{-6}\ m^3]}{[9.081 \times 10^{-8}\ m^4]} \qquad \boxed{q = 2248.21\frac{N}{m}}$$

Total Shear Load

$$F_{shear} = qL_b = \left[2248.21\frac{N}{m}\right][0.076\ m] \qquad \boxed{F_{shear} = 170.863\ N}$$

Shear Load per Screw

$$F_{screw} = \frac{F_{shear}}{4} = \frac{[170.863\ N]}{4} \qquad \boxed{F_{screw} = 42.715\ N}$$

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Avg Shear Stress per 3mm Screw

$$\tau_{avg} = \frac{4[42.715\ N]}{\pi[0.003\ m]^2} = 6.043 \times 10^6\ N/m^2$$

The yield strength for 18-8 Stainless Steel is given to be 205 Mpa from (amardeepsteel), thus the 3mm screw is strong enough to transmit the shear loads.

Answers to Problem 2:

- Shear load transmitted per screw: $F_{screw} = 42.715\ N$
- This shear load is reasonable since the average shear stress per 3mm screw in single shear is only around 6 Mpa while the yield strength for 18-8 stainless steel (a typical screw material) is 205 MPa.

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Problem 3: (25 pts) The current VTOL legs (see figure 1) are 8" length round carbon fiber tubes, with 1/2" outer diameter, 1/16" wall thickness, and constructed from 0°/90° carbon fiber. They are held fixed at the spar end, but are free to deflect and rotate at the other end. What is the maximum load that one leg can take without failing, and the likely mode of failure at that maximum load? Discuss your results.

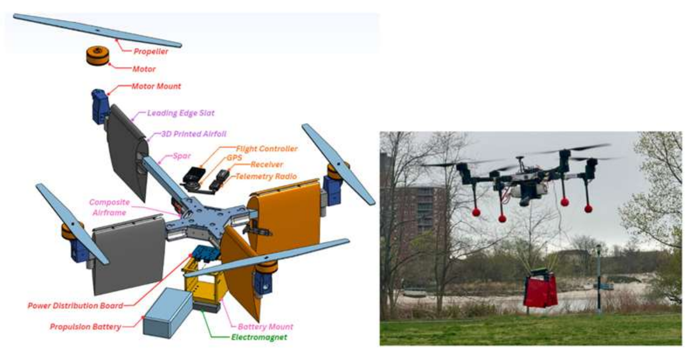

Figure 1. VFS VTOL aircraft (credit: 2025-2026 VTOL Team). (Left) Exploded view of most major components, showing three assembled spar-wing-motor-propellor assembles and one in exploded view, along with the central composite airframe body. (Right) VTOL flying with payload and legs (not shown in left figure) and with airfoils removed.

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Given:

- $d_o = 0.5\ in$ = 0.0127 m
- $t = \frac{1}{16} in$ = 0.001587 m
- 0°/90° carbon fiber: $E_{CF,0} = 40\ GPa,\ \sigma_{u,0} = 800\ MPa$
- $L = 8$ in = 0.2032 m

Approach:

- There are two modes of failure:
  - 1) Following Mohr's criterion for brittle materials, the leg will fail if the load induces a normal stress that is greater than the ultimate strength. Thus, the max load corresponding to this mode of failure is:

$$\sigma_{max} = \frac{P}{A} = \sigma_{u,0} \quad \rightarrow \quad P = A\sigma_{u,0}$$

  - 2) The leg buckles with loading. The critical loading for buckling failure is given as:

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

  For this case, the end conditions correspond to $L_e = 2L$ (one end fixed, one end free)
- The case resulting in the smaller max load will be the limiting case

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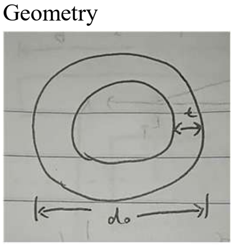
Case 1: Mohr's Criterion

$$P_1 = A\sigma_{u,0} = \left[\frac{\pi}{4}(d_o^2 - (d_o - 2t)^2)\right]\sigma_{u,0}$$

$$P_1 = \left[\frac{\pi}{4}([0.0127\text{ m}]^2 - [0.0127 - 2 \cdot 0.001587\text{ m}]^2)\right]\left[800 \times 10^6 \frac{N}{m^2}\right] \qquad \boxed{P_1 = 44324.934\ N}$$

Case 2: Buckling

Geometry



$$I = \frac{\pi}{64}[d_o^4 - (d_o - 2t)^4] = \frac{\pi}{64}\left[[0.0127\text{ m}]^4 - [0.0127 - 2 \cdot 0.001587\text{ m}]^4\right]$$

$$I = 8.727 \times 10^{-10}\ m^4$$

Critical Load

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} = \frac{\pi^2 EI}{4L^2}$$

$$P_{cr} = \frac{\pi^2[40 \times 10^9\ N/m^2][8.727 \times 10^{-10}\ m^4]}{4[0.2032\ \text{m}]^2} \qquad \boxed{P_2 = P_{cr} = 2086.014\ N}$$

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Answers to Problem 3:

- Since the max load for case 2 (buckling) is less than the max load for case 1 ($P_2 < P_1$), that means that the max load one leg can take without failing is
$$P_2 = 2086.014\ N$$
- It follows that the mode of failure would be buckling of the leg.
- Looking at the critical load formula for buckling, one way to increase the max load per leg is to change the end conditions of the leg. Namely, by adding supports to the midpoints of each leg, the effective length can be reduced (L_e), since it is squared, even a small reduction can increase the max load by a significant amount.
- Notably, the loaded aircraft weight is 89 N, so there is a comfortable safety margin. The current design should be strong enough to sustain most normal landings.